# Dheeraj Shrivastav

Smart Contract & Backend Engineer | Solidity, Foundry, TypeScript

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## SUMMARY

Smart contract and backend engineer. Founder of DappDrop, an on-chain rewards protocol (escrow, Merkle settlement, EIP-712 attestations) gated by 260+ Foundry tests; its invariant suite caught a high-severity fund drain before launch. Day job: shipping production AI agents in TypeScript/Node.js.

## PROJECTS

**DappDrop**: Sybil-resistant on-chain rewards protocol (founder) Jul 2025 - Present

*Solidity, Foundry, EIP-712, Merkle proofs, Next.js, Prisma, PostgreSQL, The Graph. Pre-launch, on Sepolia testnet.*

- Built with AI coding agents behind a verification gate I set up: invariant fuzzing, Slither/Aderyn, my review before deploy.
- Designed the protocol and security model: escrowed rewards, Merkle settlement with a 24h dispute window, fallback settler.
- Kept the core contract under the 24 KB EIP-170 limit by moving reward, NFT and fee logic into satellite modules.
- 240 unit + 26 invariant tests; an invariant caught a high-severity cross-campaign fund drain on the dev branch, pre-launch.
- Off-chain: EIP-712 attestation signer, SIWE auth, The Graph subgraph with RPC fallback, keeper worker, gasless relayer (built).

**Event Horizon**: AI-assisted grants on Base (hackathon, team of 3) Apr 2025

*Solidity, Foundry, Base, Groq, TypeScript, PostgreSQL*

- A Groq-hosted LLM scores grant proposals; a GrantDistribution contract with Foundry tests in CI pays approved ones.

## EXPERIENCE

**Product Developer**, Seragen Biotherapeutics Jun 2026 - Present

*Gemini Live, Claude, Node.js, TypeScript, WebSockets, Azure*

- Built a real-time, multilingual voice AI agent on Gemini Live (native speech-to-speech) for patients and doctors, with strict guardrails over a curated dataset; cut manual physician intervention by 65%.
- Designed a doctor-in-the-loop flow: AI drafts personalized patient reports and doctors approve them before delivery.
- Built internal AI agents that give staff on-demand recommendations, cutting routine escalations to seniors.

**Software Engineer Intern**, Scalefull Technologies Feb 2026 - May 2026

- Improved backend performance by ~25% with caching layers and REST API refactors (Node.js, TypeScript, NestJS).
- Designed and migrated PostgreSQL schemas to speed up queries on high-concurrency read paths.
- Raised test coverage on core microservices to ~90% and standardized API documentation with Swagger/OpenAPI.

**Freelance Software Developer**, Self-employed Oct 2023 - Sep 2025

- Built and deployed 10+ web applications for clients (React, Next.js, Node.js, PostgreSQL), actively used by 500+ users.
- Roughly halved API response times with Redis caching and query optimization in Express/Prisma services.
- Set up Docker-based CI/CD with GitHub Actions, replacing manual deploys.

## OPEN SOURCE

**jam-nodes** (wespreadjam), external workflow-automation framework, 213 GitHub stars Mar 2026

- 5 merged: Dev.to integration (27 tests), WordPress integration, webhook trigger, sliding-window rate limiter, DataForSEO ops.

## SKILLS

**Web3:** Foundry (unit, fuzz, invariant), Slither, Aderyn, ethers.js, viem, EIP-712, Merkle proofs, The Graph

**Languages:** TypeScript, JavaScript, Python, Solidity, SQL

**AI:** Gemini Live, Claude, AWS Bedrock, Groq, voice agents, AI agents, guardrails, MCP

**Backend:** Node.js, NestJS, Express, Bun, REST, Socket.io, Prisma, PostgreSQL, Redis, OAuth, SIWE

**Frontend:** React, Next.js, Tailwind CSS, Phaser

**Infra:** Docker, GitHub Actions, AWS (S3, CloudFront, Lambda, Bedrock), Linux, Vercel

## EDUCATION

**B.Tech, Computer Science & Engineering (AI & ML)**, DIEMS, Maharashtra 2023 - 2026

**Diploma, Computer Science & Engineering**, MGM Polytechnic, Maharashtra 2020 - 2023

Course: Cyfrin Updraft, Advanced Smart Contract Development with Foundry (2025)